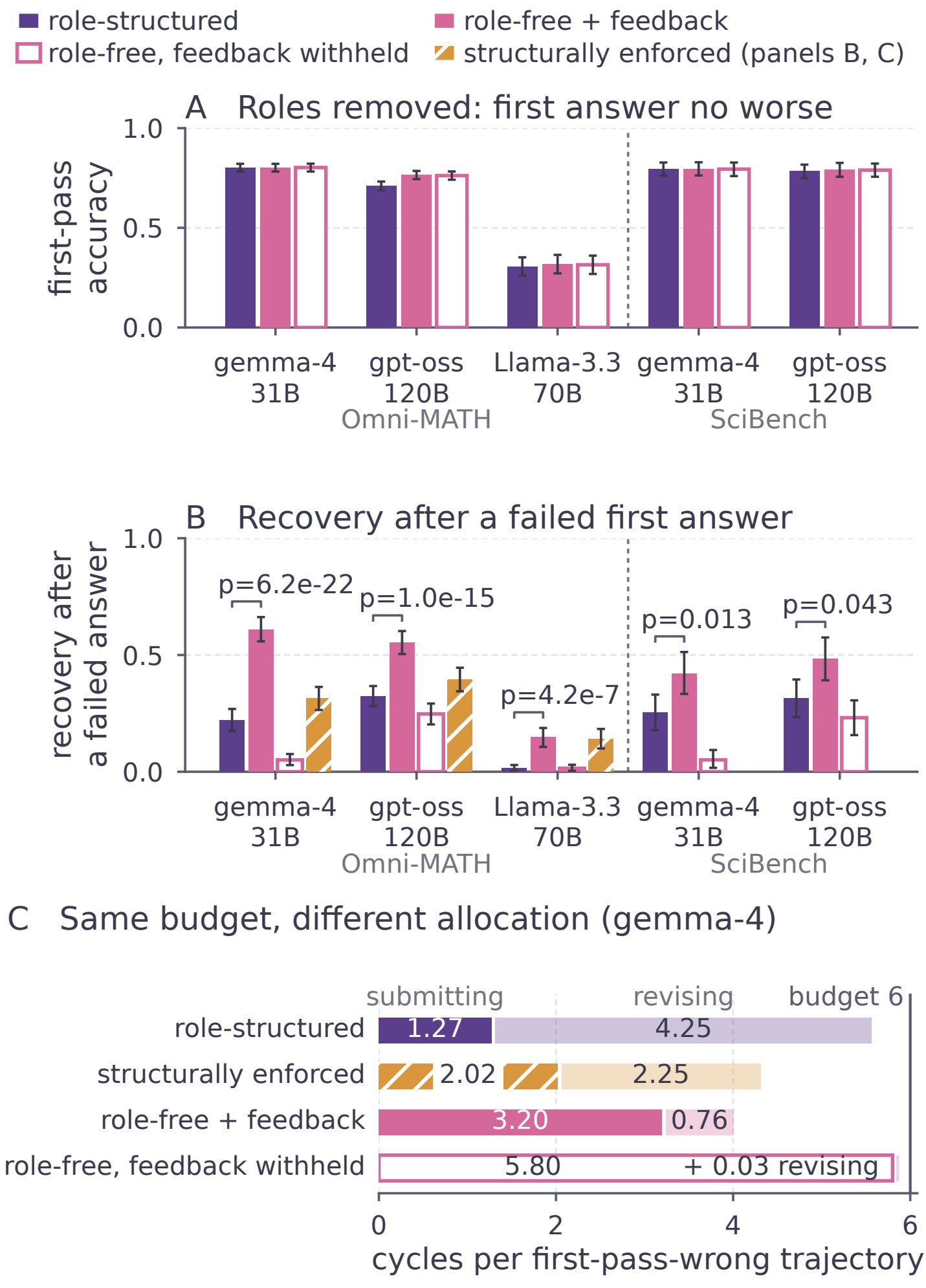
role-structured
role-free + feedback
role-free, feedback withheld
structurally enforced (panels B, C)
A Roles removed: first answer no worse
first-pass accuracy
1.0
0.5
0.0
gemma-4 31B
gpt-oss 120B
Llama-3.3 70B
gemma-4 31B
gpt-oss 120B
Omni-MATH
SciBench
B Recovery after a failed first answer
recovery after a failed answer
p=6.2e-22
p=1.0e-15
p=4.2e-7
p=0.013
p=0.043
C Same budget, different allocation (gemma-4)
submitting
revising
budget 6
role-structured
1.27
4.25
structurally enforced
2.02
2.25
role-free + feedback
3.20
0.76
role-free, feedback withheld
5.80
+ 0.03 revising
0
2
4
6
cycles per first-pass-wrong trajectory